Smart Inventory Demand Forecasting Decision Support System: An Integrated Forecast-to-Decision Prototype

1st Kavyashri NP
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
kavyashri.np@cmrit.ac.in

2nd Pavan Raj B M
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
pabm23cs@cmrit.ac.in

3rd Pavan BV
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
pabv23cs@cmrit.ac.in

4th Ranjith Bhandari
Department of Computer Science and Engineering
CMR Institute of Technology
Bengaluru, India
rabh23cs@cmrit.ac.in

***Abstract*—Inventory planning requires balancing product availability against the financial and operational risks associated with excess stock. Conventional manual estimation and static reorder rules may become inadequate when demand, selling price, inventory position, and supplier lead time vary simultaneously. This paper presents a modular web-based Decision Support System (DSS) that integrates supervised demand forecasting with deterministic inventory-risk calculations, scenario analysis, approximate model interpretation, and a human-in-the-loop procurement workflow.**

The implemented prototype uses a chronological baseline dataset containing 48 observations covering four products over twelve months. The forecasting formulation uses product identity together with previous-period sales, inventory, price, month, and supplier lead time as predictors of demand. Four supervised regression models—Linear Regression, Ridge Regression, Decision Tree Regression, and Random Forest Regression—are evaluated using a chronological 75:25 train-test split, producing 36 training observations and 12 holdout observations. Ridge Regression achieves the lowest reported Mean Absolute Error (MAE) of 13.39 and Mean Absolute Percentage Error (MAPE) of 1.70%, while Linear Regression obtains the highest reported coefficient of determination, with $R^2 = 0.9935$. Ridge Regression is selected according to the predefined primary MAE criterion.

The forecasting output is subsequently translated into operational indicators using daily demand, Safety Stock, and Reorder Point calculations. A scenario-analysis module allows users to evaluate changes in price, lead time, and demand assumptions without modifying persistent records. The system also provides empirical uncertainty estimates and baseline-perturbation sensitivity analysis, while procurement recommendations remain subject to human approval through an auditable approve/reject/snooze workflow.

The reported results should be interpreted as preliminary prototype evidence rather than broad evidence of forecasting generalization because the experiment contains only 48 observations and a single 12-observation holdout. The principal contribution is therefore the integration of forecasting, inventory-risk reasoning, scenario analysis, and auditable human decision-making within a unified DSS architecture. Larger datasets, rolling-origin validation, calibrated uncertainty estimation, and cost-aware inventory evaluation are required for stronger empirical validation.

***Index Terms*—Demand forecasting, inventory management, decision support system, machine learning, supply chain analytics, Ridge regression, safety stock, reorder point, human-in-the-loop.**

I. Introduction

Inventory management requires organizations to maintain sufficient stock to satisfy expected demand while limiting the financial and operational consequences of excessive inventory. Underestimating demand can result in stockouts, delayed fulfillment, lost sales, and service degradation, whereas overestimating demand can increase working-capital requirements, holding costs, and the risk of product obsolescence. Demand forecasting is therefore an important component of inventory planning, but forecasting alone does not determine when or how much inventory should be replenished [1], [2].

Machine-learning methods have increasingly been investigated for demand forecasting because they can represent relationships between demand and multiple operational variables. Recent reviews show that demand forecasting is one of the most widely studied machine-learning applications within supply-chain management, while inventory management and operational decision support remain closely related application areas [12], [13]. However, the predictive model is only one component of an operational inventory system. Forecasts must be transformed into actionable information involving inventory position, supplier lead time, safety stock, reorder thresholds, and procurement decisions.

Recent research has demonstrated several approaches for connecting forecasting with inventory management. Sukolkit *et al.*

integrate demand forecasting with inventory strategies for steel-wire-mesh procurement [3]. Seyedan *et al.* investigate order-up-to inventory optimization using ensemble deep-learning forecasts and forecast-demand distributions [9]. Tadayonrad and Ndiaye explicitly emphasize the relationship between demand forecasting, safety stock, supply reliability, seasonality, and inventory-related performance [6]. More recently, van der Haar *et al.* study supervised learning for integrated forecasting and inventory control, showing that forecasting and inventory decisions can be considered jointly [5].

Despite these developments, there remains a practical systems challenge: forecasting models, inventory calculations, scenario analysis, explanations, and managerial approval are frequently treated as separate components. A forecasting model may provide an accurate point estimate while offering insufficient information about uncertainty or the operational consequences of a forecast. Conversely, a conventional inventory system may provide reorder rules without exploiting predictive information from a machine-learning model. Recent work also highlights the importance of interpretability, data quality, integration, and operational deployment when machine learning is introduced into supply-chain environments [4], [12].

This paper addresses this systems-integration problem through a modular web-based DSS. The system does not propose a new machine-learning algorithm. Instead, it integrates established regression models with an inventory decision layer, scenario simulation, approximate feature-sensitivity analysis, and an auditable human-in-the-loop (HITL) approval mechanism. The objective is to demonstrate how predictive outputs can be transformed into structured decision evidence while retaining human authority over procurement actions.

The principal contributions are as follows:

1) A modular forecasting pipeline that compares multiple supervised regression models using a chronological evaluation procedure.
2) Integration of demand forecasts with deterministic Safety Stock and Reorder Point calculations.
3) Empirical uncertainty estimates and model-specific baseline-perturbation sensitivity information presented alongside point forecasts.
4) A scenario-analysis mechanism for evaluating alternative price, lead-time, and demand assumptions without modifying persistent operational records.
5) A human-in-the-loop procurement workflow that records approval decisions and contextual reasoning for auditability.
6) A decoupled software architecture connecting a web frontend, Node.js backend, Python-based machine-learning inference, persistence, and inventory decision logic.

The remainder of this paper is organized as follows. Section II reviews related literature. Section III defines the problem and research objectives. Section IV presents the system architecture. Section V describes the dataset, preprocessing, forecasting methodology, evaluation metrics, and inventory formulations. Section VI presents the experimental setup and results. Section VII describes the decision-support functionality. Section VIII discusses limitations and threats to validity. Section IX presents future research directions, followed by the conclusion in Section X.

II. Related Work

A. Machine Learning for Demand Forecasting

Machine-learning and deep-learning methods have become increasingly common in supply-chain demand forecasting. Douaioui *et al.* provide a critical review of machine-learning and deep-learning forecasting models and emphasize that model selection should reflect data characteristics and operational requirements [1]. Goel *et al.* examine machine-learning methods across different demand-forecasting horizons and highlight the importance of selecting forecasting techniques according to the forecasting context [2]. Babai *et al.* systematically review machine-learning applications in supply-chain management and report substantial growth in demand-forecasting applications, while also identifying inventory management and other operational functions as important areas for further development [12].

Sukolkit *et al.* compare several forecasting approaches for steel-wire-mesh demand and subsequently use the forecasts to support inventory planning [3]. An *et al.* investigate machine learning for demand forecasting in micro-fulfillment centers and explicitly consider interpretability and changing operational conditions [4]. Lingkon *et al.* propose a hybrid multi-item forecasting framework for complex supply-chain environments and evaluate the operational implications of improved forecasts for replenishment decisions [7].

These studies demonstrate the value of machine learning for demand prediction, but they also illustrate an important methodological point: forecasting performance cannot be interpreted independently from the characteristics of the data, evaluation procedure, and downstream operational use.

B. Forecasting and Inventory Integration

Forecasting and inventory control are closely coupled because replenishment decisions depend on both expected demand and uncertainty during the replenishment lead time. Tadayonrad and Ndiaye propose a forecasting-performance framework that explicitly considers supply reliability, seasonality, safety stock, and inventory-related outcomes [6]. Seyedan *et al.* integrate time-series demand forecasting with an order-up-to inventory optimization model and use forecast-demand distributions to estimate safety stock [9].

Van der Haar *et al.* investigate supervised learning for integrated forecasting and inventory control, demonstrating that feature-driven demand information can be incorporated directly into inventory decisions [5]. Theodorou *et al.* develop a data-driven, model-independent framework for inventory-control optimization using machine-learning models to approximate inventory performance [10]. Ren *et al.* address inventory policies under sequentially observed non-stationary demand and show that data-driven policies can outperform conventional heuristics in their experimental environments [11].

These works indicate that an important research direction lies beyond point forecasting alone: forecasts should support decisions that explicitly account for inventory position, lead time, uncertainty, and operational objectives.

C. Interpretability and Decision Support

Interpretability is particularly important in decision-support systems because users may need to understand why a model generated a particular recommendation. Lundberg and Lee introduced SHAP as a unified framework for additive feature attribution [15]. The present system does not implement SHAP; instead, it uses a simpler baseline-perturbation procedure to estimate model-specific sensitivity. This distinction is maintained explicitly to avoid overstating the explanatory guarantees of the implemented approach.

Tian and Zhang investigate a data-driven newsvendor setting in which forecasting and inventory decisions are connected through a cost-aware objective [8]. Such work reinforces the importance of considering the operational consequences of forecasts rather than evaluating predictive error in isolation.

D. Research Gap

The literature establishes strong evidence for machine-learning demand forecasting and growing evidence for integrated inventory control. However, the present work focuses on a different level of contribution. Rather than proposing a new forecasting algorithm, it combines established regression models with four operational capabilities in one web-based workflow:

1) forecast-to-inventory translation,
2) empirical uncertainty presentation,
3) interactive scenario analysis, and
4) auditable human approval of procurement recommendations.

Table I summarizes the relationship between representative studies and the proposed DSS.

III. Problem Statement and Objectives

A. Problem Statement

Retail and warehouse environments must make replenishment decisions under uncertain future demand. A static reorder threshold does not directly account for changing demand drivers such as historical sales, selling price, supplier lead time, and product-specific demand behavior. Conversely, a forecasting model that only produces a point prediction does not directly answer whether current inventory is sufficient or whether a replenishment action should be considered.

The problem addressed in this work is therefore formulated as:

> *How can a lightweight machine-learning forecasting pipeline be integrated with inventory-risk calculations and an auditable decision-support workflow so that forecast information can assist, rather than replace, managerial procurement decisions?*

The system is designed as a decision-support prototype rather than an autonomous procurement optimizer.

B. Objectives

The technical objectives are:

1) construct a modular web application for inventory and demand information;
2) compare multiple supervised regression models using chronological evaluation;
3) estimate future demand using product identity, previous-period sales, inventory, price, month, and supplier lead time;
4) integrate serialized machine-learning models with the backend inference process;
5) provide forecast-error information and empirical uncertainty estimates;
6) calculate Safety Stock and Reorder Point indicators;
7) provide interactive scenario analysis;
8) generate replenishment recommendations when inventory falls below the calculated threshold; and
9) retain human authority through an auditable approval workflow.

IV. System Architecture and Software Engineering

The proposed DSS follows a modular architecture separating presentation, application control, persistence, machine-learning inference, and inventory decision logic. The architecture is deliberately lightweight because the current implementation is a prototype intended to demonstrate end-to-end integration rather than production-scale deployment.

A. Architecture Overview

Fig. 1 presents the logical data flow.

B. Frontend Presentation Layer

The frontend is the primary interaction boundary for inventory managers. It presents inventory status, forecast information, risk indicators, recommendations, and scenario-analysis controls. Interactive charts are used to communicate historical demand and forecast information.

C. Backend API Layer

The backend is implemented using Node.js and Express. It acts as the application orchestrator and coordinates authentication, data access, forecasting requests, inventory calculations, scenario processing, and audit operations. JSON Web Tokens are used to protect authenticated routes.

D. Persistence Layer

The current development implementation uses a localized JSON-based persistence mechanism. The prototype maintains records corresponding to users, inventory states, supplier information, and audit actions.

TABLE I
COMPARISON OF REPRESENTATIVE LITERATURE AND THE PROPOSED SYSTEM

Study	Forecasting	Inventory Integration	Uncertainty	Scenario Analysis	Interpretability	HITL Audit
Douaioui et al. [1]	Yes	Partial	–	–	Partial	–
Goel et al. [2]	Yes	Partial	–	–	Partial	–
Sukolkit et al. [3]	Yes	Yes	Partial	–	–	–
Seyedan et al. [9]	Yes	Yes	Yes	–	–	–
Tadayonrad and Ndiaye [6]	Yes	Yes	Yes	–	–	–
van der Haar et al. [5]	Yes	Yes	–	–	–	–
An et al. [4]	Yes	Operational	Partial	–	Yes	–
Lingkon et al. [7]	Yes	Yes	Partial	–	Partial	–
Theodorou et al. [10]	Partial	Yes	Partial	–	–	–
Ren et al. [11]	Partial	Yes	Yes	–	–	–
Proposed DSS	**Yes**	**Yes**	**Yes**	**Yes**	**Yes***	**Yes**

*Interpretability is implemented using baseline perturbation rather than formal SHAP attribution.

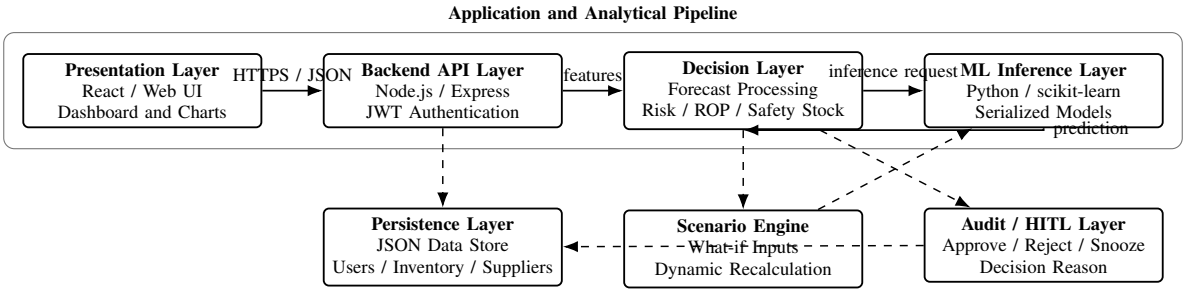


Fig. 1. Logical architecture of the proposed Smart Inventory Demand Forecasting Decision Support System.

E. Machine-Learning Inference Layer

The forecasting pipeline is implemented in Python using pandas and scikit-learn. Trained models are serialized locally and invoked by the Node.js backend through a child-process mechanism. The backend supplies the feature vector to the Python inference process and receives the numerical prediction for downstream processing.

F. Decision-Support Layer

The decision-support layer receives the forecast and combines it with current inventory and supplier lead-time information. It calculates daily demand, Safety Stock, Reorder Point, and a prototype replenishment quantity. The resulting recommendation is not automatically executed. Instead, it is placed into a pending decision state for human review.

V. METHODOLOGY

A. Dataset Description

The reported experiment uses a baseline dataset containing 48 chronological observations covering four products over twelve months. The products represented in the baseline experiment are Laptop Pro X1, Wireless Headphones, Smart Watch S3, and Gaming Mouse.

The forecasting target is unit demand. The feature representation contains product identity and five operational predictors:

- **Product ID**: categorical product identity represented numerically for model input;
- **Sales**: previous-period sales;
- **Inventory**: end-of-period inventory;
- **Price**: unit selling price;
- **Month**: chronological month indicator;
- **Lead Time**: supplier delivery duration in days.

The temporal relationship is:

$$X_t = [P_t,\, S_{t-1},\, I_t,\, Prc_t,\, M_t,\, L_t] \tag{1}$$

where S_{t-1} represents sales from the previous period and the target is:

$$y_t = D_t. \tag{2}$$

Treating sales as previous-period information is important because using contemporaneous sales that directly reflect target-period demand could introduce target leakage.

TABLE II
FEATURE DICTIONARY

Feature	Type	Operational Meaning
Demand	Target	Actual unit demand for target period
Product ID	Categorical	Product/SKU identity
Sales	Integer	Previous-period units sold
Inventory	Integer	End-of-period on-hand stock
Price	Numeric	Current selling price
Month	Temporal	Chronological month indicator
Lead Time	Integer	Supplier delivery duration in days

B. Data Quality and Preprocessing

The preprocessing stage checks schema consistency, missing values, duplicate observations, and numerical outliers. The reported baseline experiment identifies 12 price observations using the Interquartile Range (IQR) procedure:

$$IQR = Q_3 - Q_1. \tag{3}$$

An observation is flagged when:

$$x < Q_1 - 1.5IQR \quad \text{or} \quad x > Q_3 + 1.5IQR. \tag{4}$$

The identified price observations are retained because they are treated as potentially meaningful pricing variations rather than automatically discarded as erroneous measurements.

For models requiring feature scaling:

$$x_{\text{scaled}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}. \tag{5}$$

The scaling parameters are calculated using the training partition only.

C. Chronological Train-Test Procedure

Random shuffling is inappropriate for a forecasting problem because future observations may influence the training process. The reported baseline experiment therefore uses a chronological 75:25 split.

For each product:

- first nine months: training period;
- final three months: holdout period.

Thus:

$$N_{\text{train}} = 36, \qquad N_{\text{test}} = 12. \tag{6}$$

This evaluation procedure preserves temporal order, although a single holdout window is insufficient to establish broad temporal generalization. Rolling-origin validation is therefore identified as an important extension [14].

D. Regression Models

Four regression algorithms are evaluated.

TABLE III
MODEL CONFIGURATIONS

Model	Configuration
Linear Regression	fit_intercept = True
Ridge Regression	$\alpha = 1.0$, solver = auto
Decision Tree	max_depth = 5
Random Forest	100 trees, bootstrap aggregation

1) Linear Regression: The ordinary least-squares objective is:

$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2. \tag{7}$$

2) Ridge Regression: Ridge Regression introduces an L_2 regularization term:

$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \alpha \|\mathbf{w}\|_2^2, \tag{8}$$

where $\alpha = 1.0$ in the reported experiment.

3) Decision Tree Regression: Decision Tree Regression recursively partitions the feature space. The reported implementation constrains the maximum tree depth to five.

4) Random Forest Regression: Random Forest combines multiple decision trees using bootstrap aggregation. With K trees:

$$\hat{y} = \frac{1}{K} \sum_{k=1}^{K} f_k(\mathbf{x}), \tag{9}$$

where $K = 100$.

E. Evaluation Metrics

Mean Absolute Error:

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{10}$$

Root Mean Squared Error:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}. \tag{11}$$

Mean Absolute Percentage Error:

$$MAPE = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \tag{12}$$

The coefficient of determination is:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}. \tag{13}$$

MAE is used as the primary model-selection criterion.

VI. Uncertainty and Inventory Decision Formulation

A. Empirical Forecast Uncertainty

For Random Forest, the dispersion of individual-tree predictions is used as an empirical uncertainty measure:

$$\sigma_{\text{trees}} = SD\left(f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_K(\mathbf{x})\right). \tag{14}$$

A normal-approximation uncertainty band is:

$$U_{90} = \hat{y} \pm 1.645\sigma_{\text{trees}}. \tag{15}$$

This is an empirical uncertainty estimate and is not claimed to be a calibrated 90% prediction interval.

B. Baseline-Perturbation Sensitivity

The prototype does not use SHAP. Instead, model-specific feature sensitivity is estimated through baseline perturbation.

Let $\mathbf{x}_0$ denote a baseline feature vector and $\mathbf{x}^{(j)}$ denote the vector obtained by replacing feature j with its active-instance value:

$$\Delta_j = f(\mathbf{x}^{(j)}) - f(\mathbf{x}_0). \tag{16}$$

These values are interpreted as sensitivity estimates rather than exact Shapley attributions [15].

C. Daily Demand

The monthly forecast $\hat{D}$ is converted into an approximate daily rate:

$$\bar{d} = \frac{\hat{D}}{30}. \tag{17}$$

D. Safety Stock

For a nominal 95% service level:

$$Z = 1.65. \tag{18}$$

Assuming negligible supplier lead-time variability:

$$SS = Z\sigma_d\sqrt{\bar{L}}. \tag{19}$$

E. Reorder Point

The Reorder Point is:

$$ROP = \bar{d}\bar{L} + SS. \tag{20}$$

F. Prototype Order Quantity

The prototype uses:

$$Q = \max\left(0,\, \hat{D} + SS - I\right). \tag{21}$$

This is a deterministic prototype replenishment heuristic and is not an Economic Order Quantity or cost-minimizing inventory policy.

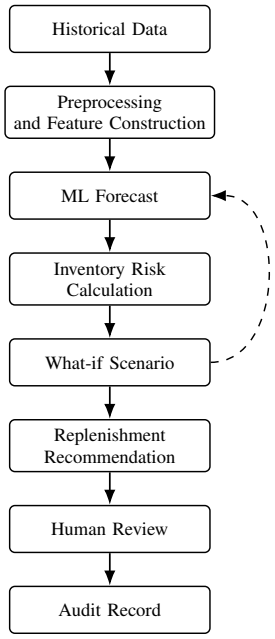


Fig. 2. Operational workflow from historical data to forecast, inventory-risk assessment, scenario analysis, recommendation, and human approval.

VII. End-to-End Decision-Support Workflow

The operational workflow consists of six stages:

1) **Data acquisition:** historical sales, inventory, price, product, month, and lead-time information are loaded.
2) **Forecast generation:** the selected regression model produces the demand estimate.
3) **Risk calculation:** forecast demand is converted into daily demand, Safety Stock, and Reorder Point indicators.
4) **Scenario analysis:** users can alter selected assumptions and request a new forecast without permanently changing operational records.
5) **Recommendation generation:** if inventory is below the calculated threshold, a replenishment recommendation is generated.
6) **Human approval and audit:** the authorized user chooses APPROVE, REJECT, or SNOOZE.

A. Human-in-the-Loop Control

The HITL design intentionally avoids fully autonomous procurement. The algorithmic recommendation is treated as decision evidence rather than as an unconditional purchase instruction.

For every action, the audit record contains the timestamp, user identifier, selected model, forecast value, decision status, and contextual justification.

VIII. Experimental Setup and Results

A. Computational Environment

The reported prototype was evaluated using an Intel Core i7 processor with 16 GB RAM. The software environment consisted of Node.js 18.x, Python 3.10, scikit-learn 1.2, and pandas 2.0.

TABLE IV
REGRESSION MODEL PERFORMANCE ON THE 12-OBSERVATION CHRONOLOGICAL HOLDOUT

Model	MAE	RMSE	MAPE (%)	R^2
Random Forest	48.10	60.30	5.97	0.9140
Linear Regression	13.87	16.58	2.00	0.9935
Ridge Regression	**13.39**	17.70	**1.70**	0.9926
Decision Tree	44.72	54.34	5.58	0.9304

The experiments are intended to validate the implemented prototype pipeline rather than establish production-scale performance.

B. Baseline Results

Table IV presents the reported performance on the 12-observation chronological holdout set.

Ridge Regression obtains the lowest MAE and MAPE and is consequently selected according to the predefined MAE criterion. Linear Regression achieves a slightly lower RMSE and higher R^2. Therefore, Ridge is selected because MAE is the primary criterion rather than because it dominates every metric.

The large performance difference between the linear models and tree-based models should be interpreted cautiously. With only 36 training observations, non-parametric tree models have limited opportunity to learn complex patterns without fitting idiosyncrasies of the training sample.

C. Interpretation of High R^2

The reported R^2 values above 0.99 require particular caution. A high coefficient of determination on a 12-observation holdout does not by itself establish generalizable forecasting ability.

Several explanations are possible:

1) strong structural relationships in the baseline data;
2) a highly informative previous-period sales variable;
3) product-specific patterns captured by Product ID;
4) chronological structure represented by Month;
5) limited diversity in the holdout observations;
6) the small sample size; or
7) potential target leakage if the temporal construction is not implemented exactly as specified.

A rigorous future validation should include feature-ablation experiments and a naive previous-period baseline.

D. Model Selection

The model-selection rule is:

$$m^* = \arg\min_m MAE_m. \tag{22}$$

Using the reported results:

$$m^* = \text{Ridge Regression}. \tag{23}$$

This selection should not be interpreted as evidence that Ridge is universally superior.

IX. DECISION-SUPPORT FEATURES

A. Scenario Simulation

Supply-chain conditions can change after a forecast has been generated. A procurement manager may want to investigate the consequences of:

- a price increase or decrease;
- a longer supplier lead time;
- a change in expected demand;
- a different inventory position.

The scenario module allows selected inputs to be modified temporarily. The backend constructs a new feature vector, invokes the forecasting process, and recalculates the downstream inventory indicators.

This design separates *analysis* from *execution*.

B. Recommendation Generation

The decision engine compares current inventory I with the calculated Reorder Point:

$$Decision = \begin{cases} \text{REPLENISH}, & I < ROP, \\ \text{MONITOR}, & I \geq ROP. \end{cases} \tag{24}$$

When replenishment is indicated, the prototype calculates Q using the deterministic rule described previously.

The recommendation contains:

- forecast demand;
- current inventory;
- Safety Stock;
- Reorder Point;
- supplier lead time;
- recommended quantity;
- uncertainty information; and
- model-sensitivity information.

C. Human Approval Workflow

The procurement queue supports three actions:

1) **APPROVE**: accept the recommendation;
2) **REJECT**: decline the recommendation and provide a reason;
3) **SNOOZE**: postpone the recommendation and provide a reason.

The workflow preserves managerial authority and therefore supports a human-in-the-loop model rather than fully autonomous procurement.

X. DISCUSSION

The experimental results provide preliminary evidence that a lightweight forecasting pipeline can be connected successfully to an inventory decision-support workflow. The primary result is not the discovery of a new regression algorithm. Instead, the system demonstrates how established predictive methods can be embedded within a broader operational workflow.

The Ridge model achieves the lowest MAE on the reported holdout set. However, the extremely high R^2 values demonstrate why model performance should not be evaluated through a single metric.

The literature provides evidence that forecasting and inventory control should be considered jointly [5], [6], [9]. The proposed system differs in emphasis by implementing a lightweight web-based decision-support workflow that combines forecast outputs, risk indicators, scenarios, sensitivity information, and human approval.

The system's decision-support architecture addresses a practical deployment issue. A forecast without context may be difficult for an inventory manager to act upon, while an autonomous purchase recommendation may be inappropriate when contextual information is unavailable to the model. The HITL workflow provides a middle ground in which the algorithm produces structured evidence and the manager retains authority over the final procurement action.

Nevertheless, the current experimental design does not demonstrate that the system reduces actual inventory cost, stockout frequency, or excess stock. It demonstrates the computational pathway from demand prediction to recommendation.

XI. Limitations and Threats to Validity

Several limitations constrain the strength of the reported conclusions.

A. Dataset Size

The baseline dataset contains only 48 observations across four products and twelve months. The 12-observation holdout is therefore too small to establish strong generalization across different market conditions, seasonal patterns, product lifecycles, or economic environments.

B. Single Holdout Window

The experiment uses one chronological holdout. Although this is preferable to random shuffling for temporal data, the reported metrics may be sensitive to the specific final three months selected for testing. Rolling-origin or expanding-window validation should be used in future experiments [14].

C. Dataset Provenance

The baseline dataset is described as a provided experimental dataset. The present evaluation does not claim that it represents a statistically representative commercial population. If the dataset is synthetic or manually constructed, its provenance should be explicitly disclosed in any external submission.

D. Potential Feature Dependence

Previous-period sales may be highly informative for future demand. This is operationally reasonable when it is genuinely available before the target period, but the forecasting result should be tested using feature-ablation experiments.

E. Inventory Simplification

The Safety Stock and Reorder Point formulations use simplified assumptions. In particular, supplier lead-time variability is treated as negligible in the implemented formulation.

F. Prototype Replenishment Rule

The order-quantity formulation is a deterministic heuristic. It does not optimize total inventory cost and does not explicitly incorporate holding, shortage, ordering, or transportation costs.

G. Uncertainty Calibration

The Random Forest uncertainty measure is based on tree-prediction dispersion and is not statistically calibrated. It should therefore not be interpreted as a guaranteed 90% prediction interval.

H. Explainability

The feature-contribution mechanism is baseline perturbation rather than SHAP. It does not possess the formal additive attribution properties of SHAP [15].

I. Persistence and Scalability

The current JSON-based persistence layer is appropriate for development and demonstration but does not establish production-scale database performance, concurrency behavior, or distributed deployment capability.

XII. Future Research Directions

The most important future improvement is expansion of the dataset. Multi-year SKU-level data would allow stronger analysis of seasonality, demand shifts, promotions, stockouts, and product-specific behavior.

Second, evaluation should move from a single chronological holdout to rolling-origin validation. This would produce multiple out-of-sample evaluation windows and reduce dependence on one particular terminal test period [14].

Third, the forecasting evaluation should include meaningful baseline methods, such as previous-period demand and appropriate moving-average or seasonal-naive methods.

Fourth, ablation experiments should quantify the contribution of Product ID, previous-period Sales, Month, Price, Inventory, and Lead Time.

Fifth, inventory evaluation should move beyond forecast error. Future experiments should measure:

- stockout rate;
- service level;
- average inventory;
- holding cost;
- shortage cost;
- inventory turnover;
- order frequency; and
- total replenishment cost.

Sixth, the uncertainty layer could be replaced or complemented by calibrated probabilistic forecasting or conformal prediction.

Seventh, the prototype persistence layer could be migrated to PostgreSQL or another transactional database when multi-user deployment is required.

Finally, future work may compare the current regression models with sequence-based approaches such as LSTM or GRU, but only after obtaining sufficient temporal data.

XIII. Conclusion

This paper presented a Smart Inventory Demand Forecasting Decision Support System that integrates supervised demand forecasting with inventory-risk calculations, scenario analysis, approximate feature sensitivity, and a human-in-the-loop procurement workflow.

The reported prototype evaluates Linear Regression, Ridge Regression, Decision Tree Regression, and Random Forest Regression on a 48-observation chronological dataset. On the 12-observation holdout, Ridge Regression obtains the lowest MAE of 13.39 and MAPE of 1.70%, while Linear Regression obtains a slightly higher R^2 of 0.9935. Ridge is selected according to the predefined MAE-based model-selection criterion.

The results should not be interpreted as evidence of broad commercial forecasting accuracy. The small dataset, single holdout window, and unusually high linear-model R^2 values require additional investigation through larger datasets, rolling-origin validation, feature ablation, and stronger baselines.

The primary contribution is therefore architectural and decision-support oriented. The system demonstrates how a forecasting model can be connected to Safety Stock and Reorder Point calculations, scenario analysis, uncertainty information, and an auditable human approval process.

The next stage of research should focus on empirical validation of inventory outcomes rather than simply increasing model complexity. Larger real-world datasets, stronger temporal validation, calibrated uncertainty estimates, cost-aware replenishment policies, and explicit service-level evaluation would provide the evidence required to move from a prototype decision-support system toward a more rigorous operational research contribution.

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